ZEROS OF THE EXTREMAL FUNCTIONS FOR POINT EVALUATION IN PALEY–WIENER SPACES

ØISTEIN SØVIK

ABSTRACT. For $0 < p < \infty$, let φ be a function of minimal L^p norm on the real line among the functions in the Paley–Wiener space PW^p with $\varphi(0) = 1$. We prove four conjectures of Brevig, Chirre, Ortega-Cerdà and Seip on the zeros of such extremal functions. For every $p > 0$ and every extremal function, the zeros are simple and the zero set is uniformly discrete; previously this was known for $p \geq 1/2$. As anticipated by these authors, it follows that $|\varphi|^{p-2}\varphi$ is a tempered distribution which reproduces the value at the origin of every Schwartz function in PW^2, now for all $0 < p < \infty$. Next, the zero set of φ together with the origin is not a uniqueness set for PW^p when $0 < p < 1$, since $z\varphi(z)$ belongs to PW^p; combined with our earlier treatment of $p \geq 1$, this settles their conjecture that it is a uniqueness set if and only if $p \geq 1$. Finally, under the uniqueness of the extremal function for small p, which is the standing assumption of the conjecture, we prove that p times the smallest positive zero has a positive and finite limit as $p \to 0^+$.

1. INTRODUCTION

For $0 < p < \infty$ let PW^p denote the Paley–Wiener space of entire functions of exponential type at most π whose restrictions to $\mathbb{R}$ belong to $L^p(\mathbb{R})$, equipped with

$$\|f\|_p^p = \int_{-\infty}^{\infty} |f(x)|^p \, \mathrm{d}x.$$

Following Brevig, Chirre, Ortega-Cerdà and Seip [3], whose results we cite with the numbering of arXiv:2210.13922v4, we let $\mathscr{C}_p$ denote the smallest constant such that $|f(0)|^p \leq \mathscr{C}_p \|f\|_p^p$ for every f in PW^p. Equivalently,

$$\frac{1}{\mathscr{C}_p} = \inf\{\|f\|_p^p : \ f \in PW^p, \ f(0) = 1\}. \tag{1}$$

This is [3, (1.6)]. A compactness argument shows that (1) has a solution for every $0 < p < \infty$, and every solution has exponential type exactly π [3, Section 1]. For $1 \leq p < \infty$ the solution is unique, and it is denoted by φ_p [3, Lemma 3.5]. For $0 < p < 1$ uniqueness is open [3, Conjecture 9.4]. We

Date: October 13, 2026.

2020 *Mathematics Subject Classification.* Primary 30D15; Secondary 30C15, 41A44, 42A38.

Key words and phrases. Paley–Wiener spaces, point evaluation, extremal functions, zeros, uniformly discrete sets, uniqueness sets.

write $\mathscr{Z}(\varphi)$ for the zero set of an entire function φ and, for a set Λ of real numbers,

$$\sigma(\Lambda) = \inf\{|\lambda - \mu| : \ \lambda, \mu \in \Lambda, \ \lambda \neq \mu\}.$$

The set Λ is *uniformly discrete* if $\sigma(\Lambda) > 0$.

Brevig, Chirre, Ortega-Cerdà and Seip studied the solutions of (1) in detail [3]. They proved that the zeros of every solution are real, that they are simple when $p \geq 1/2$ and have multiplicity at most $1/p$ when $p < 1/2$ [3, Lemma 3.2], that the zero set is a finite union of uniformly discrete sets for every p, and that it is uniformly discrete when $p \geq 1/2$ [3, Theorem 1.4]. Setting $\Phi = |\varphi|^{p-2}\varphi/\|\varphi\|_p^p$, they also proved that

$$f(0) = \int_{-\infty}^{\infty} f(x)\Phi(x)\,\mathrm{d}x \tag{2}$$

for every f in PW^p when $p \geq 1$ [3, Theorem 3.6], and that Φ is a tempered distribution and (2) holds for every f in $\mathcal{S} \cap PW^2$ when $1/2 \leq p < 1$ [3, Theorem 8.1]. Here $\mathcal{S}$ is the Schwartz class. In the final section of [3] they listed a number of conjectures, among them the following two.

Conjecture ([3, Conjecture 9.6])**.** *The zero set of any solution of* (1) *is uniformly discrete for all* $0 < p < \infty$.

Conjecture ([3, Conjecture 9.7])**.** *Theorem 8.1 of* [3] *extends to the full range* $0 < p < \infty$.

The authors explain that the first conjecture is singled out because it implies the second. The main purpose of the present paper is to prove both.

Theorem 1. *Suppose that* $0 < p < \infty$ *and that* φ *is a solution of* (1)*. Then* $\mathscr{Z}(\varphi)$ *is uniformly discrete.*

Theorem 2. *Suppose that* $0 < p < \infty$ *and that* φ *is a solution of* (1)*. Then every zero of* φ *is simple.*

Theorem 3. *Suppose that* $0 < p < \infty$ *and that* φ *is a solution of* (1)*. Then* Φ *is a tempered distribution, and* (2) *holds for every* f *in* $\mathcal{S} \cap PW^2$.

The cases $p \geq 1/2$ of these three theorems are in [3], as recalled above. Theorem 3 is deduced from Theorems 1 and 2. Separation alone does not suffice for it: at a zero of multiplicity $m \geq 2$ with $mp \leq 1$, the function $|\varphi|^{p-1}$ is not locally integrable, since $m(1-p) \geq 1$.

The remaining two conjectures concern the extremal function itself, so for $0 < p < 1$ they are stated under the uniqueness assumption. Following [3] we write $0 < t_1(p) < t_2(p) < \cdots$ for the positive zeros of φ_p.

Conjecture ([3, Conjecture 9.10])**.** *We have* $\lim_{p\to\infty} t_1(p) = 1/2$ *and* $\lim_{p\to 0^+} p\, t_1(p) > 0$.

Conjecture ([3, Conjecture 9.11])**.** $\mathscr{Z}(\varphi_p) \cup \{0\}$ *is a uniqueness set for* PW^p *if and only if* $p \geq 1$.

Here a set $\Lambda \subset \mathbb{C}$ is a *uniqueness set* for PW^p if the only function in PW^p which vanishes on Λ is the zero function. The conjectures are motivated by the case $p = 2$, where $\varphi_2(z) = \operatorname{sinc} \pi z$ with $\operatorname{sinc} z = (\sin z)/z$, so that $x\varphi_2(x) = \pi^{-1} \sin \pi x$ and $\mathscr{Z}(\varphi_2) \cup \{0\}$ is the set of integers. In a previous paper [6, Theorems 2 and 3] we proved the first half of [3, Conjecture 9.10] and the "if" direction of [3, Conjecture 9.11]; the case $p \geq 2$ of the latter was observed in [3], and the case $p = 1$ is also contained in work of Bondarenko, Ortega-Cerdà, Radchenko and Seip [1,2]. Here we prove the remaining halves. The first does not need uniqueness.

Theorem 4. *Suppose that $0 < p < 1$ and that φ is a solution of (1). Then $z\varphi(z)$ is in PW^p. In particular, $\mathscr{Z}(\varphi) \cup \{0\}$ is not a uniqueness set for PW^p.*

Combined with [6, Theorem 3], Theorem 4 proves [3, Conjecture 9.11]. The second half of [3, Conjecture 9.10] is stated after the sentence "We now take the uniqueness of the extremal function for granted" in [3, Section 9.3], and we prove it under that assumption, which we only need for small p.

Theorem 5. *Suppose that there is $p_0 > 0$ such that (1) has a unique solution φ_p for every $0 < p < p_0$. Then the limit $\lim_{p \to 0^+} p\, t_1(p)$ exists, and it is positive and finite.*

Combined with [6, Theorem 2], Theorem 5 proves [3, Conjecture 9.10] as posed. To the best of our knowledge, Theorems 1 to 5 are new in the stated ranges.

Let us briefly describe our approach. The proof of Theorem 1 starts from the orthogonality relation

$$\int_{-\infty}^{\infty} \frac{x^2 |\varphi(x)|^p}{(x-a)(x-b)} \, \mathrm{d}x = 0 \tag{3}$$

for distinct zeros a and b of φ [3, Lemma 3.3]. Only the interval (a, b) contributes negatively. When $b - a$ is small and $p < 1/2$, the negative and the positive contributions near (a, b) have the same order of magnitude, which is why the argument of [3] requires $p \geq 1/2$. We remove every zero in a small cluster around (a, b), approximate what remains by an exponential on the scale of the cluster, and show by a contour integral of $\prod_j (z - r_j)^p/(z^2 - 1)$ that for every monic polynomial P with roots r_j in $[-1, 1]$, including ± 1, the positive part of $|P|^p/(u^2 - 1)$ outweighs the negative part by a fixed factor, uniformly in the exponential weight. Theorem 2 follows by splitting a multiple zero into two nearby zeros; the change of the norm is a beta integral, which turns out to be negative. With Theorems 1 and 2 in hand, the proof of [3, Lemma 8.3] gives a lower bound for $|\varphi|$, and Theorem 3 follows from it, as anticipated in [3]. For Theorem 4, the competitor $\varphi(\lambda z) \operatorname{sinc}(\pi(1 - \lambda)z)$ shows that the tail $\int_{|x| \geq R} |\varphi|^p$ is $O(1/R)$ for every p; when $p < 1$ this is enough to sum the weight $|x|^p$ over dyadic shells. For Theorem 5 we rescale, $w_p(\tau) = |\varphi_p(\tau/p)|^p$, and show that w_p converges in $L^1(\mathbb{R})$ as $p \to 0^+$, by

testing (1) with products of dilated extremal functions. Moving the first zero shows that every limit point of $p\,t_1(p)$ minimizes a logarithmic energy of the limit profile. The limit points form an interval, since $p\,t_1(p)$ is continuous under the uniqueness assumption, and this energy is real analytic below the largest limit point; hence there is only one limit point.

Organization. The present paper is organized into six additional sections. In Section 2 we collect the facts from [3] and the classical results that we use, together with two lemmas shared by several of the proofs. Section 3 contains the proof of Theorem 1, and Section 4 the proofs of Theorems 2 and 3. In Section 5 we prove Theorem 4, and in Section 6 we prove Theorem 5. Section 7 contains some concluding remarks.

2. Preliminaries

Throughout, φ denotes a solution of (1) for some fixed $0 < p < \infty$, unless stated otherwise. Note that $\|\varphi\|_p^p = 1/\mathscr{C}_p$. We will use the following results from [3].

- (B1) The function φ is real on $\mathbb{R}$ [3, Lemma 2.3]. The zeros of φ are real and have multiplicity at most $\max(1, 1/p)$ [3, Lemma 3.2].
- (B2) $\mathscr{Z}(\varphi)$ is a finite union of uniformly discrete sets [3, Theorem 1.4].
- (B3) If a and b are distinct zeros of φ, then (3) holds, and the integral converges absolutely [3, Lemmas 3.1 and 3.3].
- (B4) If f is a real function in PW^∞ and $|f(\xi)| = \|f\|_\infty$, then $|f(x)| \geq \|f\|_\infty \cos \pi(x - \xi)$ for $|x - \xi| \leq 1/2$ [3, Lemma 3.4].
- (B5) If f is an even function in PW^∞ with $\|f\|_\infty = f(0) = 1$, and $0 < t_1 \leq t_2 \leq \cdots$ are its positive zeros repeated according to multiplicity, then $t_n \geq n/e$ for every n [3, Lemma 7.2].
- (B6) $\lim_{p\to 0^+} 2\mathscr{C}_p/p = c_0$ for a constant $0 < c_0 < \infty$ [3, Theorem 1.3].

Here PW^∞ is the space of entire functions of exponential type at most π which are bounded on $\mathbb{R}$.

Classical facts. By a theorem of Plancherel and Pólya [5] (see also [4, Lecture 7]), a function of exponential type which is in $L^p(\mathbb{R})$ is bounded on $\mathbb{R}$. Hence φ is in PW^∞. Applying $|f(0)|^p \leq \mathscr{C}_p \|f\|_p^p$ to the translates $f = \varphi(\cdot + x)$, we find that

$$|\varphi(x)| \leq 1 = \varphi(0), \qquad x \in \mathbb{R}. \tag{4}$$

By the Phragmén–Lindelöf principle, $|\varphi(x + iy)| \leq e^{\pi|y|}$, and Bernstein's inequality gives $|\varphi'(x)| \leq \pi$ for real x [4, Lecture 28].

Since φ is bounded on $\mathbb{R}$, it is in the Cartwright class. As φ is real on $\mathbb{R}$ and has exponential type exactly π, its indicator diagram is the segment $[-i\pi, i\pi]$. By the theorem on the distribution of zeros of functions in the Cartwright class [4, Lecture 17], the positive zeros $t_1 \leq t_2 \leq \cdots$ of φ, repeated according to multiplicity, satisfy $t_n/n \to 1$, and so do the absolute values of the negative zeros. In particular, the number of zeros in an interval of length 1 is bounded,

and $\sum_t t^{-2} < \infty$. Since φ has exponential type and $\varphi(0) = 1$, Hadamard's factorization theorem gives

$$\varphi(z) = e^{az} \prod_{t \in \mathscr{Z}(\varphi)} \left(1 - \frac{z}{t}\right)^{m(t)} e^{m(t)z/t},$$

where $m(t)$ is the multiplicity of the zero t, and $a = \varphi'(0)$ is real. The first lemma records the consequence of this formula that we need. It is the identity (4.11) in the proof of [3, Lemma 4.2], applied after a polynomial factor has been removed; we include the short proof since we use it for functions with multiple zeros. In words, $\log|Q|$ is concave between consecutive zeros of Q, and its second derivative is small where the zeros of Q are far away.

Lemma 1. *Suppose that P is a real polynomial whose zeros, counted with multiplicity, are zeros of φ. Then $Q = \varphi/P$ is entire, and*

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} \log|Q(x)| = - \sum_{t \in \mathscr{Z}(Q)} \frac{m_Q(t)}{(x-t)^2}$$

for every real x outside $\mathscr{Z}(Q)$, where $m_Q(t)$ is the multiplicity of t as a zero of Q.

Proof. Each factor $z - r$ of P is a constant times $1 - z/r$, since $\varphi(0) \neq 0$. Removing these factors from the Hadamard product and absorbing the corresponding factors $e^{z/r}$ into the exponential, we get $Q(z) = Be^{a'z} \prod_t (1 - z/t)^{m_Q(t)} e^{m_Q(t)z/t}$ with $B \neq 0$ and a' real. For real x outside $\mathscr{Z}(Q)$, the second derivative of $\log|1 - x/t|$ is $-(x-t)^{-2}$, and that of $a'x + x/t$ vanishes. The series of second derivatives converges locally uniformly on $\mathbb{R} \setminus \mathscr{Z}(Q)$, since $\sum_t t^{-2} < \infty$, so it may be differentiated term by term. □

The second lemma is a competitor which is used twice: to bound the tails of φ in Section 5, and to compare $\mathscr{C}_q$ with $\mathscr{C}_p$ for nearby exponents in Section 6. The idea is to dilate φ slightly, to $\varphi(\lambda z)$, which costs a factor $1/\lambda$ in the norm and frees the exponential type $(1 - \lambda)\pi$. The freed type is spent on a power of a sinc function, which leaves the values near the origin almost unchanged and damps the tails. Comparing the gain in the tails with the cost of the dilation gives (a).

Lemma 2. *Suppose that $0 < p < \infty$ and that φ is a solution of (1).*

(a) *For every $R > 0$,*

$$\int_{|x| \geq R} |\varphi(x)|^p \, \mathrm{d}x \leq \frac{2}{\mathscr{C}_p (\pi R + 2)(1 - 2^{-p})}.$$

(b) $\limsup_{q \to p} 1/\mathscr{C}_q \leq 1/\mathscr{C}_p$.

Proof. For $0 < \lambda < 1$ and an integer $N \geq 1$ put

$$f(z) = \varphi(\lambda z) \left(\operatorname{sinc} \frac{\pi(1-\lambda)z}{N}\right)^N.$$

Then f is entire of exponential type at most $\lambda\pi+(1-\lambda)\pi=\pi$, $f(0)=1$, and $|f(x)|\le|\varphi(\lambda x)|$ for real x, since $|\operatorname{sinc} t|\le 1$. Hence f is in PW^q whenever $\varphi(\lambda\,\cdot)$ is in $L^q(\mathbb{R})$.

(a) Take $N=1$ and $q=p$. Since f competes in (1), the substitution $y=\lambda x$ gives

$$\frac{1}{\mathscr{C}_p}\le\|f\|_p^p=\frac{1}{\lambda}\int_{\mathbb{R}}|\varphi(y)|^p|\operatorname{sinc}(by)|^p\,\mathrm{d}y,\qquad b=\frac{\pi(1-\lambda)}{\lambda}.$$

Since $\lambda=\pi/(\pi+b)$, this rearranges to

$$\int_{\mathbb{R}}|\varphi(y)|^p\bigl(1-|\operatorname{sinc}(by)|^p\bigr)\,\mathrm{d}y\le\frac{1-\lambda}{\mathscr{C}_p}=\frac{b}{(\pi+b)\mathscr{C}_p}.$$

Every $b>0$ arises from some $0<\lambda<1$, and we take $b=2/R$. The integrand is nonnegative, and for $|y|\ge R$ we have $|\operatorname{sinc}(by)|\le 1/(b|y|)\le 1/2$, so $1-|\operatorname{sinc}(by)|^p\ge 1-2^{-p}$ there. Since $b/(\pi+b)=2/(\pi R+2)$, this proves (a).

(b) Let $0<\delta<1$, put $\lambda=1-\delta$, and choose an integer $N>2/p$. By (4), $|f(x)|\le 1$, and $|f(x)|\le(N/(\pi\delta|x|))^N$. Hence $|f|^q\le|f|^{p/2}$ for $q\ge p/2$, and $|f|^{p/2}$ is integrable since $Np/2>1$. In particular f is in PW^q and $1/\mathscr{C}_q\le\|f\|_q^q$ for $q\ge p/2$. By dominated convergence,

$$\limsup_{q\to p}\frac{1}{\mathscr{C}_q}\le\int_{\mathbb{R}}|f(x)|^p\,\mathrm{d}x\le\int_{\mathbb{R}}|\varphi((1-\delta)x)|^p\,\mathrm{d}x=\frac{1}{(1-\delta)\mathscr{C}_p}.$$

Letting $\delta\to 0^+$, we obtain (b). □

We will also use competitors of the form φr, where r is a real rational function with $r(0)=1$ which is bounded outside a compact set, and whose poles are zeros of φ of at least the same multiplicity. Such a function φr is entire, has exponential type at most π, is in $L^p(\mathbb{R})$, and takes the value 1 at the origin, so $\|\varphi\|_p^p\le\|\varphi r\|_p^p$.

3. Uniform separation of the zeros

Let us first explain why zeros could cluster at all when $p<1/2$. Suppose that $a<b$ are simple zeros of φ with $\delta=b-a$ small, that no other zero is nearby, and let $c=(a+b)/2$. Near c we have $|\varphi(x)|\approx A|(x-a)(x-b)|$ for some $A>0$, so the absolute value of the integrand of (3) is approximately $c^2A^p|(x-a)(x-b)|^{p-1}$. The negative contribution, from (a,b), is therefore of order $c^2A^p\delta^{2p-1}$. The positive contribution from $\delta<|x-c|<L$ is of order $c^2A^p\int_\delta^L u^{2p-2}\,\mathrm{d}u$. If $p\ge 1/2$, this is of order $c^2A^pL^{2p-1}$ or $c^2A^p\log(L/\delta)$, which is much larger than the negative contribution when δ is small, and the orthogonality relation fails. If $p<1/2$, however, both contributions are of order $c^2A^p\delta^{2p-1}$ and both come from the scale δ itself. Hence no comparison of orders of magnitude decides the sign, and the two parts must be compared exactly on the scale of the zeros.

This exact comparison requires three steps. First, a and b may be accompanied by further zeros on the same scale, possibly multiple ones. We

therefore enlarge the pair to a cluster of k zeros whose distance to all other zeros is much larger than its diameter, and we rescale the cluster to $[-1,1]$. Second, the remaining factor Q of φ must be replaced by something explicit. By Lemma 1, the second derivative of $\log|Q|$ is small near the cluster, since the zeros of Q are far away. Hence $\log|Q|$ is affine on the scale of the cluster up to a small error, and after rescaling $|Q|^p$ is a constant times e^{su}. The slope s is not under control, so the comparison must hold uniformly in s. Third, we need the model inequality for $|P(u)|^p e^{su}/(u^2-1)$, where P is the rescaled polynomial; this is the next lemma.

The model inequality comes from a contour integral. The function $F(z) = \prod_j (z-r_j)^p$ is holomorphic in the upper half-plane, and the argument of its boundary value at a real point u is πp times the number of roots to the right of u. Outside $[-1,1]$ this argument takes its extreme values 0 and πpk, while inside $(-1,1)$ it lies strictly between them, because ± 1 are roots. Cauchy's theorem gives $\int_{\mathbb{R}} F(u+i0)(u^2-1)^{-1}\,\mathrm{d}u = 0$, which is a model of the orthogonality relation (3). After a rotation by $e^{-i\pi kp/2}$, the real part of $F(u+i0)$ is a smaller fraction of $|P(u)|^p$ for $|u|>1$ than for $|u|<1$. Since the two parts balance, the positive part $\int_{|u|>1}|P|^p/(u^2-1)$ must exceed the negative part by a fixed factor. When $kp \geq 1$, which includes $k=2$ and $p \geq 1/2$, the positive part is infinite, in accordance with the heuristic above.

Lemma 3. *Suppose that $p>0$ and that $k \geq 2$ is an integer. There are $M>2$ and $\varepsilon>0$, depending only on p and k, such that every polynomial $P(u) = \prod_{j=1}^k (u-r_j)$ with $r_j \in [-1,1]$ for every j and with both -1 and 1 among the r_j satisfies*

$$\int_{1<|u|<M} e^{su} \frac{|P(u)|^p}{u^2-1}\,\mathrm{d}u \geq (1+\varepsilon)\int_{-1}^{1} e^{su}\frac{|P(u)|^p}{1-u^2}\,\mathrm{d}u \tag{5}$$

for every real s. Repeated roots are allowed.

Proof. Write $q(u) = |P(u)|^p/|u^2-1|$ and $I = \int_{-1}^1 q(u)\,\mathrm{d}u$. We first bound I from above and below, uniformly in P. Since $|P(u)| \leq 2^{k-2}(1-u^2)$ for $|u| \leq 1$, we have

$$I \leq C \coloneqq 2^{(k-2)p}\int_{-1}^{1}(1-u^2)^{p-1}\,\mathrm{d}u < \infty.$$

For the lower bound, remove from $[-1/2,1/2]$ the intervals of radius $1/(4k)$ around the roots. The remaining set has length at least $1/2$, and on it each factor of P has modulus at least $1/(4k)$. Hence $I \geq a \coloneqq (4k)^{-kp}/2$.

Suppose first that $kp<1$. In the upper half-plane define $F(z) = \prod_j (z-r_j)^p$ with the branches $0<\arg(z-r_j)<\pi$. The function $F(z)/(z^2-1)$ is holomorphic in the upper half-plane and is $O(|z|^{kp-2})$ at infinity. Near a root r of multiplicity m its modulus is $O(|z-r|^{mp-1})$ if $r=\pm 1$ and $O(|z-r|^{mp})$ otherwise. By Cauchy's theorem applied to a large half-disc indented by

small half-circles above ± 1, we get

$$\int_{-\infty}^{\infty} \frac{F(u+i0)}{u^2-1}\,\mathrm{d}u = 0,$$

since the large arc contributes $O(R^{kp-1})$ and a small half-circle of radius η contributes $O(\eta^{mp})$. For real u which is not a root, let $j(u)$ be the number of roots larger than u, counted with multiplicity. Then $F(u+i0) = |P(u)|^p e^{i\pi p j(u)}$. We multiply the identity by $e^{-i\pi kp/2}$ and take real parts. For $|u| > 1$ we have $j(u) \in \{0, k\}$, and the real part of $e^{i\pi p(j(u)-k/2)}$ is $\cos(\pi kp/2) > 0$. For $|u| < 1$ we have $1 \le j(u) \le k-1$, so $|j(u)-k/2| \le k/2-1$ and the real part is at least $\cos(\pi p(k-2)/2)$. Since $u^2 - 1 < 0$ on $(-1,1)$, we find that

$$\int_{|u|>1} q(u)\,\mathrm{d}u \ge \gamma I, \qquad \gamma \coloneqq \frac{\cos(\pi p(k-2)/2)}{\cos(\pi pk/2)} > 1.$$

Since $|P(u)| \le (|u|+1)^k$, the integral of q over $|u| \ge M_0$ is $O(M_0^{kp-1})$ uniformly in P for $M_0 \ge 2$. Choose M_0 so that this integral is at most $(\gamma-1)a/2$. Since $a \le I$, we obtain

$$\int_{1<|u|<M_0} q(u)\,\mathrm{d}u \ge \gamma_0 I, \qquad \gamma_0 \coloneqq \frac{1+\gamma}{2} > 1.$$

If $kp \ge 1$, then $q(u) \ge (u-1)^{kp}/(u^2-1)$ for $u \ge 2$, and the integral of the right-hand side over $[2,\infty)$ diverges. Choose M_0 such that its integral over $[2, M_0]$ is at least $2C$. Then the last display holds with $\gamma_0 = 2$.

It remains to insert the weight e^{su}. When $|s|$ is small, the weight is almost constant on $(-M_0, M_0)$. When $|s|$ is large, the weight is much larger at one end of $(-M, M)$ than on $[-1,1]$, and q is not small there. Let $s_0 = \log\gamma_0/(2(M_0+1))$. If $|s| \le s_0$, then $e^{su} \ge e^{-|s|M_0}$ for $|u| < M_0$ and $e^{su} \le e^{|s|}$ for $|u| \le 1$, so the ratio of the left-hand side of (5), with M_0 in place of M, to the integral on the right-hand side is at least $\gamma_0 e^{-|s|(M_0+1)} \ge \sqrt{\gamma_0}$. If $|s| > s_0$, we may by symmetry assume that $s > 0$, and we use only the interval $[M-1, M]$. For $M \ge 3$ and u in this interval, $q(u) \ge (M-2)^{kp} M^{-2}$, so the left-hand side of (5) is at least $(M-2)^{kp} M^{-2} e^{s(M-1)}$, while the integral on the right-hand side is at most Ce^s. Choose $M \ge \max(M_0, 3)$ so large that $(M-2)^{kp} M^{-2} e^{s_0(M-2)} \ge 2C$. Then the ratio is at least 2 when $s > s_0$. Hence (5) holds with $\varepsilon = \min(\sqrt{\gamma_0} - 1, 1)$. □

Armed with Lemma 3, we carry out the three steps described above.

Proof of Theorem 1. We argue by contradiction. Let $K \ge 2$ be an integer such that every interval of length 1 contains at most K zeros of φ, counted with multiplicity; it exists by (B1) and (B2). Suppose that there are pairs of distinct zeros $a_j' < b_j'$ of φ with $\delta_j = b_j' - a_j' \to 0$. Since a nonzero entire function has only finitely many zeros in a compact set, the pairs escape to infinity.

Step 1: isolated clusters. Let $R_j = \delta_j^{-1/(2K)}$. This tends to infinity, but slowly enough that the cluster built below stays small. Start with the cluster consisting of the two zeros a_j' and b_j'. If a zero of φ outside the cluster has distance at most R_j times the diameter of the cluster from the cluster, we add it to the cluster, and we repeat. Each addition multiplies the diameter by at most $1 + R_j \le 2R_j$. If $K-1$ additions were possible, the cluster would contain at least $K+1$ distinct zeros and have diameter at most $\delta_j (2R_j)^{K-1} \le 2^K \sqrt{\delta_j}$, which is less than 1 for large j. This contradicts the choice of K. Hence the process stops, and for large j we obtain a cluster with diameter at most $2^K\sqrt{\delta_j}$ whose distance to every other zero exceeds R_j times its diameter. We count the zeros in the cluster with multiplicity; passing to a subsequence, we may assume that their number is a fixed integer $2 \le k \le K$. Write $a_j = c_j - h_j$ and $b_j = c_j + h_j$ for the smallest and the largest zero in the cluster, and d_j for the distance from c_j to the nearest zero outside the cluster. Then $h_j \to 0$, $|c_j| \to \infty$ and $d_j/h_j \to \infty$.

Step 2: the remaining factor. Write

$$\varphi(z) = Q_j(z) \prod_{i=1}^{k} (z - c_j - h_j r_{j,i}), \qquad r_{j,i} \in [-1,1],$$

where the product runs over the zeros in the cluster, counted with multiplicity. Both -1 and 1 occur among the $r_{j,i}$. Fix M and ε as in Lemma 3. For large j we have $Mh_j < d_j/2$, so Q_j has no zeros in $|x - c_j| \le Mh_j$, and every zero of Q_j has distance at least $d_j/2$ from this interval. By Lemma 1,

$$\bigl|(\log|Q_j|)''(x)\bigr| = \sum_{t \in \mathscr{Z}(Q_j)} \frac{m_{Q_j}(t)}{(x-t)^2} \le C_K(1 + d_j^{-2})$$

for $|x - c_j| \le Mh_j$. Indeed, at most $4K$ zeros lie within distance 2 of x, and each contributes at most $4/d_j^2$, while the zeros in $[x+n, x+n+1]$ and in $[x-n-1, x-n]$, $n \ge 2$, contribute at most $2K/n^2$. Taylor's formula therefore gives

$$p \log \frac{|Q_j(c_j + h_j u)|}{|Q_j(c_j)|} = s_j u + o(1), \qquad s_j = p h_j (\log|Q_j|)'(c_j),$$

uniformly for $|u| \le M$, because $h_j^2(1 + d_j^{-2}) \to 0$. No bound on s_j is needed.

Step 3: the contradiction. Let $P_j(u) = \prod_i (u - r_{j,i})$, and apply (3) to the zeros a_j and b_j. In the substitution $x = c_j + h_j u$ we have $(x - a_j)(x - b_j) = h_j^2(u^2 - 1)$ and $|\varphi(x)|^p = h_j^{kp} |Q_j(x)|^p |P_j(u)|^p$. Hence the absolute value of the integrand of (3) on the interval $|u| \le M$, including $\mathrm{d}x$, equals

$$|Q_j(c_j)|^p h_j^{kp-1} c_j^2 \, e^{s_j u} \, \frac{|P_j(u)|^p}{|u^2 - 1|} \, \theta_j(u) \, \mathrm{d}u, \qquad \sup_{|u| \le M} |\theta_j(u) - 1| \to 0,$$

where θ_j accounts for the error term of Step 2 and for the factor $(1 + h_j u/c_j)^2$. This is a relative error, which does not depend on the size of s_j or of $|Q_j(c_j)|$. The integrand of (3) is negative exactly on (a_j, b_j), which corresponds to

$|u| < 1$, and it is nonnegative elsewhere. By Lemma 3 with $s = s_j$, the integral over $1 < |u| < M$ alone exceeds the absolute value of the integral over $|u| < 1$ by a factor at least $(1+\varepsilon)(1-o(1))/(1+o(1))$, which is larger than 1 for large j. This contradicts (3). □

4. Simple zeros and the reproducing formula

We begin with Theorem 2. For $p \geq 1/2$ it is [3, Lemma 3.2], so we may assume that $p < 1/2$. A multiple zero is split into two nearby zeros, and we compute the change of the norm. The competitor below replaces the factor $(z-t)^2$ of φ by $(z-t-\varepsilon)(z-t+\varepsilon)$, up to normalization. It is the competitor used in the proof of [3, Lemma 3.2] to exclude double zeros when $p = 1/2$. When $mp < 1$, the cusp $|x-t|^{mp}$ of $|\varphi|^p$ at t is so sharp that the change of the norm comes from the scale ε around t. It is of order ε^{mp+1}, which beats the cost $O(\varepsilon^2)$ of the normalization, and its sign is that of a beta integral. When $mp = 1$ we recover the logarithmic gain found in [3] for $p = 1/2$ and $m = 2$.

Proof of Theorem 2. Let $0 < p < 1/2$ and suppose that t is a zero of φ of multiplicity $m \geq 2$. Then $t \neq 0$, $mp \leq 1$ by (B1), and $\varphi(t+u) = u^m g(t+u)$ with g entire and $g(t) \neq 0$. For small $\varepsilon > 0$, the competitor

$$\varphi_\varepsilon(z) = \frac{1 - \varepsilon^2/(z-t)^2}{1 - \varepsilon^2/t^2}\,\varphi(z)$$

is of the form discussed at the end of Section 2, so $\|\varphi\|_p^p \leq \|\varphi_\varepsilon\|_p^p$. We have $\|\varphi_\varepsilon\|_p^p = (1-\varepsilon^2/t^2)^{-p}(\|\varphi\|_p^p + T_\varepsilon)$, where

$$T_\varepsilon = \int_{\mathbb{R}} |\varphi(x)|^p \left(\left| 1 - \frac{\varepsilon^2}{(x-t)^2} \right|^p - 1 \right) \mathrm{d}x.$$

Since $(1-\varepsilon^2/t^2)^{-p} = 1 + O(\varepsilon^2)$, it suffices to show that $T_\varepsilon \leq -c\,\varepsilon^2 \omega(\varepsilon)$ for some $c > 0$ and a function ω with $\omega(\varepsilon) \to \infty$ as $\varepsilon \to 0^+$.

Suppose first that $mp < 1$. Fix a small $\eta > 0$. The part of T_ε with $|x-t| \geq \eta$ is $O(\varepsilon^2)$. In the remaining part we substitute $x = t + \varepsilon u$ and use dominated convergence, to obtain

$$T_\varepsilon = |g(t)|^p \varepsilon^{mp+1} J + o(\varepsilon^{mp+1}), \qquad J = \int_{\mathbb{R}} |u|^{(m-2)p} \big(|u^2-1|^p - |u|^{2p} \big)\, \mathrm{d}u.$$

The integral converges absolutely, since its integrand is $O(|u|^{(m-2)p})$ near 0 and $O(|u|^{mp-2})$ at infinity, and this also provides the dominating function. Since $mp + 1 < 2$, it remains to show that $J < 0$. Set $\alpha = (1 + (m-2)p)/2$ and $\beta = \alpha + p = (1+mp)/2$. The substitution $v = u^2$ gives

$$\begin{aligned} J &= \int_0^1 v^{\alpha-1} \big((1-v)^p - v^p \big)\, \mathrm{d}v + \int_1^\infty v^{\alpha-1} \big((v-1)^p - v^p \big)\, \mathrm{d}v \\ &= \frac{p}{\beta} \big(\mathrm{B}(\alpha, p) - \mathrm{B}(1-\beta, p) \big), \end{aligned}$$

where B is the beta function. For the second equality, we substitute $v = 1/w$ in the second integral and integrate by parts in both integrals; the boundary terms $\pm 1/\beta$ cancel. Since $0 < 1 - \beta < \alpha$ and $\mathrm{B}(\cdot, p)$ is strictly decreasing, $J < 0$.

Suppose next that $mp = 1$. Then $|\varphi(t+y)|^p = |y||g(t+y)|^p$, and $|g|^p$ is Lipschitz near t. The part of T_ε with $|x-t| \le \varepsilon$ is $O(\varepsilon^2)$, and so is the part with $|x-t| \ge \eta$. For $\varepsilon < |x-t| < \eta$ we use $|1-w|^p = 1 - pw + O(w^2)$, uniformly for $0 \le w \le 1$. This gives

$$T_\varepsilon = -2p|g(t)|^p \varepsilon^2 \log \frac{1}{\varepsilon} + O(\varepsilon^2).$$

In both cases $\|\varphi_\varepsilon\|_p < \|\varphi\|_p$ for small ε, which is a contradiction. □

We turn to Theorem 3. For $p \ge 1$ it is [3, Theorem 3.6], since $\mathcal{S} \cap PW^2 \subset PW^p$, and Φ is in $L^{p/(p-1)}(\mathbb{R})$, or in $L^\infty(\mathbb{R})$ if $p = 1$. For $1/2 \le p < 1$ it is [3, Theorem 8.1]. For the rest of this section we fix $0 < p < 1/2$ and a solution φ of (1), and we write $\sigma = \sigma(\mathscr{Z}(\varphi))$, which is positive by Theorem 1. The next lemma is [3, Lemma 8.3], whose proof uses the assumption $p \ge 1/2$ only through the separation of the zeros, as noted in [3]. Since our exponent differs and we use Lemma 1 in place of [3, Lemma 4.2], we include the proof.

Lemma 4. *There is $C > 0$ such that $|\varphi(x)| \ge C \operatorname{dist}(x, \mathscr{Z}(\varphi))(1+|x|)^{-1-4/p}$ for every real x.*

Proof. Let $t_1 < t_2 < \cdots$ be the positive zeros of φ; they are simple by Theorem 2. Since $t_n/n \to 1$, we may choose N such that $t_N \ge 1$ and $t_{n+2} \le 2t_n$ for $n \ge N$. For $n \ge N$ put $I_n = [t_n, t_{n+1}]$,

$$\psi_n(x) = \frac{\varphi(x)}{(x-t_n)(x-t_{n+1})}, \qquad \Psi_n(x) = \frac{x^2|\varphi(x)|^p}{|(x-t_n)(x-t_{n+1})|},$$

and $M_n = \max_{I_n} |\psi_n|$. By (B3), $\int_{\mathbb{R}\setminus I_n} \Psi_n = \int_{I_n} \Psi_n$. On I_n we have $\Psi_n(x) = x^2|\psi_n(x)|^p\big((x-t_n)(t_{n+1}-x)\big)^{p-1}$. Since $2p - 1 < 0$ and $|I_n| \ge \sigma$,

$$\int_{I_n} \Psi_n(x)\,\mathrm{d}x \le M_n^p t_{n+1}^2 |I_n|^{2p-1} \mathrm{B}(p,p) \le M_n^p t_{n+1}^2 \sigma^{2p-1} \mathrm{B}(p,p).$$

By (4) and (B4) with $\xi = 0$, $|\varphi(x)| \ge \cos \pi x \ge 1/2$ for $|x| \le 1/3$. The interval $[0, 1/3]$ lies outside I_n, and there $(t_n - x)(t_{n+1} - x) \le t_{n+1}^2$, so $\int_{\mathbb{R}\setminus I_n} \Psi_n \ge 2^{-p} t_{n+1}^{-2}/81$. Hence

$$M_n \ge c_1 t_{n+1}^{-4/p}, \qquad c_1 = \left(\frac{\sigma^{1-2p}}{81 \cdot 2^p\, \mathrm{B}(p,p)}\right)^{1/p}. \tag{6}$$

Let $\xi_n \in I_n$ be a point with $|\psi_n(\xi_n)| = M_n$, and put $\tilde{\psi}_n(x) = \varphi(x)/((x - t_n)(x - t_{n+1})(x - t_{n+2}))$. The zeros of $\tilde{\psi}_n$ lie outside (t_{n-1}, t_{n+3}), where for $n = N$ we let t_{N-1} denote the largest zero of φ less than t_N. By Lemma 1, $\log|\tilde{\psi}_n|$ is concave on $[\xi_n, \xi_{n+1}] \subset [t_n, t_{n+2}]$, so its minimum there is attained at an endpoint. Since $\tilde{\psi}_n(\xi_n) = \psi_n(\xi_n)/(\xi_n - t_{n+2})$ and

$\tilde{\psi}_n(\xi_{n+1}) = \psi_{n+1}(\xi_{n+1})/(\xi_{n+1} - t_n)$, where both denominators have modulus at most t_{n+2}, (6) gives

$$|\tilde{\psi}_n(x)| \geq c_1 t_{n+2}^{-1-4/p} \geq c_1 (2x)^{-1-4/p}, \qquad \xi_n \leq x \leq \xi_{n+1},$$

where we used $t_{n+2} \leq 2t_n \leq 2x$. For such x, the zero of φ nearest to x is one of t_n, t_{n+1}, t_{n+2}, and the other two have distance at least $\sigma/2$ from x. Hence $|(x-t_n)(x-t_{n+1})(x-t_{n+2})| \geq (\sigma^2/4)\operatorname{dist}(x, \mathscr{Z}(\varphi))$, and

$$|\varphi(x)| \geq \frac{c_1 \sigma^2}{2^{3+4/p}} \operatorname{dist}(x, \mathscr{Z}(\varphi))\, x^{-1-4/p}, \qquad x \geq \xi_N,$$

since the intervals $[\xi_n, \xi_{n+1}]$, $n \geq N$, cover $[\xi_N, \infty)$. On $[0, \xi_N]$ there are finitely many zeros, all simple, so $|\varphi(x)|/\operatorname{dist}(x, \mathscr{Z}(\varphi))$ is bounded below there by a positive constant. Finally, $\varphi(-z)$ is also a solution of (1), which covers $x \leq 0$. □

Lemma 5. *There is a constant $C' > 0$ such that*

$$\int_k^{k+1} |\varphi(x)|^{p-1}\, \mathrm{d}x \leq C'(1+|k|)^{(1-p)(1+4/p)}$$

for every integer k.

Proof. By Lemma 4, $|\varphi(x)|^{p-1} \leq C^{p-1}(2+|k|)^{(1-p)(1+4/p)} \operatorname{dist}(x, \mathscr{Z}(\varphi))^{p-1}$ for $k \leq x \leq k+1$. If $\operatorname{dist}(x, \mathscr{Z}(\varphi)) < 1$ for such x, then the nearest zero lies in $(k-1, k+2)$, and this interval contains at most $3/\sigma + 1$ zeros. Hence $\int_k^{k+1} \operatorname{dist}(x, \mathscr{Z}(\varphi))^{p-1}\, \mathrm{d}x \leq 1 + (3/\sigma + 1) \cdot 2/p$. □

We are now ready to prove Theorem 3. Instead of [3, Lemma 8.2], we differentiate under the integral sign with the help of the elementary inequality

$$\bigl||a+b|^p - |a|^p - p|a|^{p-2}ab\bigr| \leq K_p |a|^{p-1}|b| \min\bigl(1, |b|/|a|\bigr), \tag{7}$$

valid for real $a \neq 0$, real b and $0 < p < 1$, with $K_p = 2^{2-p}(3^p + 2^p) + 2p$. If $|b| \leq |a|/2$, Taylor's formula applied to $s \mapsto |a+sb|^p$ on $[0,1]$, where $|a+sb| \geq |a|/2$, bounds the left-hand side by $2^{1-p}p(1-p)|a|^{p-2}b^2$. If $|b| > |a|/2$, then $|a+b|^p + |a|^p \leq (3^p + 2^p)|b|^p$ and $|b|^p < 2^{1-p}|a|^{p-1}|b|$, so the left-hand side is at most $(2^{1-p}(3^p+2^p) + p)|a|^{p-1}|b|$, while $\min(1, |b|/|a|) \geq 1/2$.

Proof of Theorem 3 for $0 < p < 1/2$. By Lemma 5, Φ is locally integrable, and $\int_{\mathbb{R}} |\Phi(x)|(1+|x|)^{-n}\, \mathrm{d}x < \infty$ for $n > (1-p)(1+4/p) + 1$. Hence Φ is a tempered distribution.

Let f be in $\mathcal{S} \cap PW^2$. Writing $f^*(z) = \overline{f(\bar{z})}$, the functions $(f+f^*)/2$ and $(f-f^*)/(2i)$ are real on $\mathbb{R}$ and belong to $\mathcal{S} \cap PW^2$, so we may assume that f is real on $\mathbb{R}$. Since f decays faster than any power, f is in PW^p. Put $g = f - f(0)\varphi$. For real ε, the function $\varphi + \varepsilon g$ is in PW^p and takes the value 1 at the origin. Hence $G(\varepsilon) = \int_{\mathbb{R}} |\varphi + \varepsilon g|^p\, \mathrm{d}x$ has a minimum at $\varepsilon = 0$. Moreover, $|\varphi|^{p-1}|g| \leq |\varphi|^{p-1}|f| + |f(0)||\varphi|^p$ is integrable, by Lemma 5

and the rapid decay of f. Applying (7) with $a = \varphi(x)$ and $b = \varepsilon g(x)$ for x outside the null set $\mathscr{Z}(\varphi)$, we get

$$\left|\frac{G(\varepsilon) - G(0)}{\varepsilon} - p\int_{\mathbb{R}} |\varphi|^{p-2}\varphi g \,\mathrm{d}x\right| \le K_p \int_{\mathbb{R}} |\varphi|^{p-1}|g| \min\bigl(1, |\varepsilon g|/|\varphi|\bigr)\,\mathrm{d}x.$$

The right-hand side tends to 0 as $\varepsilon \to 0$ by dominated convergence. Hence $G'(0) = p\int_{\mathbb{R}} |\varphi|^{p-2}\varphi g \,\mathrm{d}x$, and it vanishes since G has a minimum at 0. Since $\int_{\mathbb{R}} |\varphi|^{p-2}\varphi g \,\mathrm{d}x = \int_{\mathbb{R}} f|\varphi|^{p-2}\varphi \,\mathrm{d}x - f(0)\|\varphi\|_p^p$, this is (2). □

5. Uniqueness sets

Theorem 4 is a direct consequence of the tail bound in Lemma 2. Let us explain why $z\varphi(z)$ should be in PW^p for $p < 1$. By Lemma 2 (a), the mass of $|\varphi|^p$ on $R \le |x| < 2R$ is $O(1/R)$, so multiplying by $|x|^p \approx R^p$ gives mass $O(R^{p-1})$, which is summable over dyadic R exactly when $p < 1$. For $p = 2$ the bound $O(1/R)$ is attained, since $|\varphi_2(x)|^2 \approx x^{-2}$ on average. The theorem also agrees with the rationale given in [3, Section 9.3] for their conjecture, namely that $|x\varphi_p(x)|/\operatorname{dist}(x, \mathscr{Z}(\varphi_p) \cup \{0\})$ behaves roughly like $(1 + |x|)^{1-2/p}$, whose pth power is integrable if and only if $p < 1$.

Proof of Theorem 4. Let $0 < p < 1$ and $\psi(z) = z\varphi(z)$. Then ψ is entire, and it has exponential type at most π since φ has. For $n \ge 0$ let $E_n = \{x \in \mathbb{R} : 2^n \le |x| < 2^{n+1}\}$. By Lemma 2 (a) with $R = 2^n$,

$$\int_{E_n} |x\varphi(x)|^p \,\mathrm{d}x \le 2^{(n+1)p} \int_{|x|\ge 2^n} |\varphi(x)|^p \,\mathrm{d}x \le \frac{2^{p+1}}{\pi(1 - 2^{-p})\mathscr{C}_p}\, 2^{-n(1-p)}.$$

Since $p < 1$, the right-hand side is summable over $n \ge 0$, and ψ is bounded on $[-1, 1]$. Hence ψ is in $L^p(\mathbb{R})$, and so in PW^p. It is not identically zero, since $\varphi(0) = 1$, and it vanishes on $\mathscr{Z}(\varphi) \cup \{0\}$, with at least the multiplicity of each zero of φ. □

The same dyadic argument shows that $\int_{\mathbb{R}} |x|^\alpha |\varphi(x)|^p \,\mathrm{d}x < \infty$ for every $0 \le \alpha < 1$ and every $0 < p < \infty$. In the other direction, $x\varphi_p$ is not in $L^p(\mathbb{R})$ when $p \ge 1$, as shown in Step 3 of the proof of [6, Theorem 3]. Neither that proof nor the proofs of the present paper use [3, Theorem 8.6]; see [6, Section 7] for a remark on its formula (8.12).

6. The first zero as $p \to 0^+$

Throughout this section we assume that (1) has a unique solution φ_p for every $0 < p < p_0$, and we consider only such p. Since $\varphi_p(-z)$ is also a solution, φ_p is even. By (B1), Theorem 2 and the Hadamard factorization in Section 2,

$$\varphi_p(z) = \prod_{n=1}^{\infty} \Bigl(1 - \frac{z^2}{t_n^2}\Bigr), \qquad 0 < t_1 < t_2 < \cdots,$$

where $t_n = t_n(p)$. By (4) and (B5), $t_n \geq n/e$. For $a > 0$ and real τ put $\ell_a(\tau) = \log|1 - \tau^2/a^2|$, and define

$$w_p(\tau) = |\varphi_p(\tau/p)|^p, \qquad \mu_p = p \sum_{n \geq 1} \delta_{p t_n}, \qquad b(p) = p\, t_1(p),$$

where δ_a is the unit point mass at a. Then $0 \leq w_p \leq 1$ by (4), and $J_p \coloneqq \int_{\mathbb{R}} w_p = p/\mathscr{C}_p$ tends to $J_0 \coloneqq 2/c_0$ as $p \to 0^+$ by (B6). The product formula gives

$$\log w_p = U_{\mu_p}, \qquad U_\nu(\tau) \coloneqq \int \ell_a(\tau)\, \mathrm{d}\nu(a), \tag{8}$$

and $t_n \geq n/e$ gives $\mu_p((0,r]) \leq er$ for every $r > 0$. Every positive measure ν on $(0,\infty)$ with $\nu((0,r]) \leq er$ for every $r > 0$ satisfies

$$\int_{(A,\infty)} \frac{\mathrm{d}\nu(a)}{a^2} = \int_A^\infty \frac{2\nu((A,r])}{r^3}\, \mathrm{d}r \leq \frac{2e}{A}, \qquad A > 0, \tag{9}$$

where the identity follows from $a^{-2} = \int_a^\infty 2r^{-3}\, \mathrm{d}r$ and Fubini's theorem.

Let us explain the idea of the proof before we begin. In the variable $\tau = px$, the zeros of φ_p become the atoms of μ_p, each of mass p, and by (8) the function w_p is the exponential of the logarithmic potential of μ_p. The bound $t_n \geq n/e$ says that μ_p has density at most e, and $b(p)$ is the left end of the support of μ_p. For $p = 1/k$, the competitor $\operatorname{sinc}^{2k}(\pi x/(2k))$ used in [3, Section 7.3] corresponds to the measure with mass 2 at each point $2, 4, 6, \ldots$, whose support starts at 2. Thus the conjecture says that the support of μ_p does not reach the origin in the limit, and that its left end converges. One could try to identify the limiting variational problem for μ, whose value would be the constant c_0 of (B6), but we do not know how to solve it. Instead we use the following route. First, we prove that w_p converges in $L^1(\mathbb{R})$ to a profile W, by testing (1) with products of two dilated extremal functions, a variant of the power trick of [3, Lemma 2.1]. Second, moving the pair of zeros $\pm t_1$ to $\pm c/p$ changes $\log w_p$ by $p(\ell_c - \ell_{b(p)})$, so to first order in p it changes $\int w_p$ by $p \int W(\ell_c - \ell_{b(p)})$. Since φ_p is extremal, every limit point of $b(p)$ therefore minimizes the energy $c \mapsto \int W \ell_c$, normalized as $\mathcal{I}(c)$ below. Third, an energy which is real analytic on the relevant interval and tends to infinity at 0 cannot have an interval of minimum points.

We first record that b is continuous. This is the only place, apart from evenness, where the uniqueness assumption is used.

Lemma 6. *The function b is continuous on $(0, p_0)$.*

Proof. Fix $0 < p < p_0$ and let $q_k \to p$. By (4) and the Phragmén–Lindelöf bound, the functions φ_{q_k} are locally uniformly bounded in $\mathbb{C}$. Let φ_* be the locally uniform limit of a subsequence. Then $\varphi_*(0) = 1$ and $|\varphi_*(x+iy)| \leq e^{\pi|y|}$, and Fatou's lemma and Lemma 2 (b) give

$$\int_{\mathbb{R}} |\varphi_*(x)|^p\, \mathrm{d}x \leq \liminf_{k\to\infty} \int_{\mathbb{R}} |\varphi_{q_k}(x)|^{q_k}\, \mathrm{d}x = \liminf_{k\to\infty} \frac{1}{\mathscr{C}_{q_k}} \leq \frac{1}{\mathscr{C}_p}.$$

So φ_* solves (1), and $\varphi_* = \varphi_p$ by uniqueness. Hence $\varphi_{q_k} \to \varphi_p$ locally uniformly along the whole sequence. Let $t = t_1(p)$ and $0 < \varepsilon < t$. The function φ_p has no zeros in the disc $|z| \le t-\varepsilon$ and vanishes at t. By Hurwitz's theorem, for large k the function φ_{q_k} has no zeros in $|z| \le t - \varepsilon$ and has a zero in $|z - t| < \varepsilon$, which is real by (B1). Hence $|t_1(q_k) - t| < \varepsilon$ for large k. □

The next lemma shows that the rescaled extremal functions converge. It does not use uniqueness; any choice of solutions would do. The competitor in the proof is a product of two dilated extremal functions, in the spirit of the power trick [3, Lemma 2.1] and of the proof of [3, Corollary 7.4]. Since J_p has a limit by (B6), Hölder's inequality for this product is almost an equality, and a quantitative form of Hölder's inequality turns this into convergence of w_p.

Lemma 7. *There is a function W in $L^1(\mathbb{R})$ such that $\|w_p - W\|_{L^1} \to 0$ as $p \to 0^+$. Moreover, W is even, $0 \le W \le 1$ and $\int_{\mathbb{R}} W = J_0$.*

Proof. For $0 < \alpha < 1$ and $x, y \ge 0$ we have

$$x^\alpha y^{1-\alpha} \le \alpha x + (1-\alpha)y - \min(\alpha, 1-\alpha)\bigl(\sqrt{x} - \sqrt{y}\bigr)^2. \tag{10}$$

By symmetry we may assume that $\alpha \le 1/2$. The difference between the right-hand and the left-hand sides is $(1-2\alpha)y + 2\alpha\sqrt{xy} - x^\alpha y^{1-\alpha}$, which is nonnegative by the inequality between weighted arithmetic and geometric means, since $y^{1-2\alpha}(\sqrt{xy})^{2\alpha} = x^\alpha y^{1-\alpha}$.

Let $0 < q' \le q < p_0$, let $m \ge 1$ be the integer nearest to q/q', and define p by $1/p = m/q + 1/q'$. Then $p \le q'$. Put $\alpha = mp/q$, so that $1 - \alpha = p/q'$. Since $m/(q/q')$ lies in $[2/3, 3/2]$, we have $2/5 \le \alpha \le 3/5$. The function $f(z) = \varphi_q(pz/q)^m \varphi_{q'}(pz/q')$ has exponential type at most $\pi p(m/q + 1/q') = \pi$ and satisfies $f(0) = 1$. The substitution $\tau = px$ gives

$$p \int_{\mathbb{R}} |f(x)|^p \,\mathrm{d}x = \int_{\mathbb{R}} w_q^\alpha w_{q'}^{1-\alpha} \,\mathrm{d}\tau.$$

By Hölder's inequality the right-hand side is finite. So f is in PW^p, and $J_p \le \int w_q^\alpha w_{q'}^{1-\alpha}$. Combining this with (10), we get

$$\bigl\|\sqrt{w_q} - \sqrt{w_{q'}}\bigr\|_{L^2}^2 \le \frac{5}{2}(\alpha J_q + (1-\alpha)J_{q'} - J_p).$$

As $q, q' \to 0^+$ we also have $p \to 0^+$, so the right-hand side tends to 0 by (B6). Hence $\sqrt{w_q}$ converges in $L^2(\mathbb{R})$ to some function S. Since $\|w_q - w_{q'}\|_{L^1} \le \|\sqrt{w_q} - \sqrt{w_{q'}}\|_{L^2}\|\sqrt{w_q} + \sqrt{w_{q'}}\|_{L^2}$, the functions w_q converge in $L^1(\mathbb{R})$ to $W = S^2$. The remaining properties of W pass to the limit from those of w_q. □

The next lemma shows that the scaled first zero stays away from the origin. We divide out the first pair of zeros, which makes the competitor larger near the origin and smaller far away.

Lemma 8. *There are $\kappa > 0$ and $0 < p_1 < p_0$ such that $b(p) \geq \kappa$ for $0 < p < p_1$.*

Proof. Let $p \leq 1/2$ be so small that $J_p \geq 3J_0/4$, and put $t = t_1(p)$ and $b = pt$. The function $\psi(z) = \varphi_p(z)/(1 - z^2/t^2)$ is a competitor of the form discussed in Section 2, so $\int|\varphi_p|^p \leq \int|\psi|^p$. Since $|\psi| \leq |\varphi_p|$ for $|x| \geq \sqrt{2}t$, this gives

$$\int_{|x|<\sqrt{2}t} |\varphi_p|^p \biggl(\Bigl| 1 - \frac{x^2}{t^2} \Bigr|^{-p} - 1 \biggr) \,\mathrm{d}x \geq \int_{|x|\geq\sqrt{2}t} |\varphi_p|^p \biggl(1 - \Bigl| 1 - \frac{x^2}{t^2} \Bigr|^{-p} \biggr) \,\mathrm{d}x.$$

On the left-hand side we use $|\varphi_p| \leq 1$, $e^y - 1 \leq ye^y$ for $y \geq 0$, and $p \leq 1/2$. This bounds the left-hand side by $A_0 pt = A_0 b$, where

$$A_0 = \int_{|u|<\sqrt{2}} |1 - u^2|^{-1/2} \log \frac{1}{|1 - u^2|} \,\mathrm{d}u < \infty.$$

On the right-hand side, put $R = J_0/8$ and suppose that $b \leq R/2$. Then $R/p \geq 2t$, and for $|x| \geq R/p$ we have $|1 - x^2/t^2| \geq x^2/(2t^2) \geq R^2/(2b^2)$. With $\lambda = \log(R^2/(2b^2)) > 0$, the integrand on the right-hand side is at least $|\varphi_p|^p(1 - e^{-p\lambda})$ there. Since $w_p \leq 1$, we have $\int_{|\tau|\geq R} w_p \geq J_p - 2R \geq J_0/2$. Using $1 - e^{-y} \geq y/(1 + y)$, we obtain

$$A_0 b \geq \frac{1 - e^{-p\lambda}}{p} \int_{|\tau|\geq R} w_p(\tau) \,\mathrm{d}\tau \geq \frac{J_0}{2} \cdot \frac{\lambda}{1 + p\lambda} \geq \frac{J_0}{4} \min(\lambda, 1/p).$$

If $\lambda \leq 1/p$, this says that $A_0 b \geq (J_0/4)\log(R^2/(2b^2))$, which fails when b is smaller than a constant $\kappa_0 > 0$ depending only on J_0. If $\lambda > 1/p$, then $b \geq J_0/(4A_0 p)$, which exceeds $R/2$ for small p, contrary to the assumption $b \leq R/2$. So $b(p) \geq \kappa := \min(\kappa_0, R/2)$ for all small p. □

By Lemma 8, for $p < p_1$ the measures μ_p are supported in $[\kappa, \infty)$, and $\mu_p((0, r]) \leq er$. Hence every sequence $p_k \to 0^+$ has a subsequence along which μ_{p_k} converges vaguely on $(0, \infty)$, that is, against compactly supported continuous functions on $(0, \infty)$, to a positive measure μ supported in $[\kappa, \infty)$ with $\mu((0, r]) \leq er$ for every $r > 0$. The next lemma identifies W in terms of such a limit.

Lemma 9. *Suppose that $p_k \to 0^+$ and that $\mu_{p_k} \to \mu$ vaguely. Then $U_{\mu_{p_k}} \to U_\mu$ in $L^1_{\mathrm{loc}}(\mathbb{R})$, and $W = e^{U_\mu}$ almost everywhere.*

Proof. Fix $T > 0$ and $A \geq \max(2T, \kappa)$. Choose a continuous function $\chi\colon (0, \infty) \to [0, 1]$ with $\chi = 1$ on $[\kappa/2, A]$ and support in $(\kappa/4, A + 1)$. Let ν be any one of the measures μ_{p_k} and μ. The part $\int(1 - \chi(a))\ell_a \,\mathrm{d}\nu(a)$ involves only $a > A$. There $\tau^2/a^2 \leq 1/4$ for $|\tau| \leq T$, so $|\ell_a(\tau)| \leq 2\tau^2/a^2$, and by (9) this part is at most $4eT^2/A$ on $[-T, T]$.

For the part $\int \chi(a)\ell_a \,\mathrm{d}\nu(a)$ we truncate at height $H > 0$, putting $\ell_a^H = \max(\ell_a, -H)$. For $a < A + 1$ we have $-\ell_a(\tau) \leq -\log|a - \tau| - \log|a + \tau| +$

$2\log(A+1)$. Using also $(x+y-H)_+ \le (2x-H)_+ + (2y-H)_+$, we get

$$\int_{-T}^{T} (\ell_a^H(\tau) - \ell_a(\tau))\,\mathrm{d}\tau \le 2\int_{\mathbb{R}} \bigl(-2\log|u| + 2\log(A+1) - H\bigr)_+ \,\mathrm{d}u =: \varepsilon_H$$

for $\kappa/4 < a < A+1$, and $\varepsilon_H \to 0$ as $H \to \infty$. The truncation error in $L^1(-T,T)$ is therefore at most $e(A+1)\varepsilon_H$. Finally, the function $(a,\tau) \mapsto \chi(a)\ell_a^H(\tau)$ is continuous on $[\kappa/4, A+1] \times [-T,T]$. Hence the family of functions $a \mapsto \chi(a)\ell_a^H(\tau)$, $|\tau| \le T$, is bounded and equicontinuous, and vague convergence gives $\int \chi(a)\ell_a^H(\tau)\,\mathrm{d}\mu_{p_k}(a) \to \int \chi(a)\ell_a^H(\tau)\,\mathrm{d}\mu(a)$ uniformly for $|\tau| \le T$. Choosing first A, then H, and then k, we obtain the first claim.

By (8), $w_{p_k} = e^{U_{\mu_{p_k}}}$. Along a further subsequence, $U_{\mu_{p_k}} \to U_\mu$ almost everywhere, and $w_{p_k} \to W$ almost everywhere by Lemma 7. This proves the second claim. □

Corollary 1. $\limsup_{p\to 0^+} b(p) < \infty$.

Proof. If $b(p_k) \to \infty$, then every compact subset of $(0,\infty)$ has μ_{p_k}-measure 0 for large k, so every vague limit is $\mu = 0$. Then Lemma 9 gives $W = 1$ almost everywhere, which contradicts $\int_{\mathbb{R}} W = J_0 < \infty$. □

For $c > 0$ we define the logarithmic energy

$$\mathcal{I}(c) = \int_{\mathbb{R}} W(\tau)(\ell_c(\tau) - \ell_1(\tau))\,\mathrm{d}\tau.$$

The integral converges absolutely. Indeed, $\ell_c - \ell_1 = \log|c^2-\tau^2| - \log|1-\tau^2| - 2\log c$ is bounded for large $|\tau|$ and has only logarithmic singularities, while $0 \le W \le 1$ and W is integrable. For the same reason, $\mathcal{I}(c) - \mathcal{I}(c') = \int W(\ell_c - \ell_{c'})$. The next lemma is the variational condition obtained by moving the first pair of zeros.

Lemma 10. *If $p_k \to 0^+$ and $b(p_k) \to b_0$, then $\mathcal{I}(b_0) = \min_{c>0} \mathcal{I}(c)$.*

Proof. By Lemma 8 and Corollary 1, $0 < b_0 < \infty$. Fix $c \ne b_0$. For large k write $p = p_k$, $t = t_1(p)$ and $b = b(p)$, so that $b_0/2 \le b \le 2b_0$, and put $s = c/p$. The function $\psi(z) = \varphi_p(z)(1-z^2/s^2)/(1-z^2/t^2)$ is a competitor of the form discussed in Section 2, so $\int|\varphi_p|^p \le \int|\psi|^p$. In the variable $\tau = px$ this reads

$$0 \le \frac{1}{p}\int_{\mathbb{R}} w_p(e^{p\Delta} - 1)\,\mathrm{d}\tau, \qquad \Delta := \ell_c - \ell_b. \tag{11}$$

Let $\tilde\varphi(z) = \varphi_p(z)/(1-z^2/t^2)$. For $x \ge 0$, Bernstein's inequality gives $|\varphi_p(x)| = |\varphi_p(x) - \varphi_p(t)| \le \pi|x-t|$, hence $|\tilde\varphi(x)| \le \pi t^2/(t+x) \le \pi t$. By evenness this holds on all of $\mathbb{R}$. Therefore

$$w_p e^{p\Delta} = |\tilde\varphi(\tau/p)|^p |1-\tau^2/c^2|^p \le (2\pi b_0/p)^p |1-\tau^2/c^2|^p,$$

and $(2\pi b_0/p)^p \le 2$ for large k. Let $N = \{\tau : \ \bigl||\tau| - b\bigr| < \delta\}$ with $\delta = \min(|c-b_0|, b_0)/8$. Off N, the ratio $e^{\Delta} = |1-\tau^2/c^2|/|1-\tau^2/b^2|$ is bounded by a constant $K_1 \ge 1$ which does not depend on k, since it is continuous there and tends to b^2/c^2 at infinity. On N, $w_p e^{p\Delta_+} \le w_p + w_p e^{p\Delta} \le K_2$ for

a constant K_2. Since $e^y - 1 \le y + y^2 e^{y_+}/2$ for real y, the right-hand side of (11) is at most

$$\int_{\mathbb{R}} w_p \Delta \,\mathrm{d}\tau + \frac{p}{2}\Big(K_1 \int_{\mathbb{R}\setminus N} w_p \Delta^2 \,\mathrm{d}\tau + K_2 \int_N \Delta^2 \,\mathrm{d}\tau \Big).$$

The bracket is bounded in k. Indeed, Δ is bounded for $|\tau| \ge T$, with T independent of k, so $\int_{|\tau|\ge T} w_p \Delta^2 \le C J_p$, while on $|\tau| \le T$ we have $w_p \le 1$ and $\Delta^2 \le 2\ell_c^2 + 2\ell_b^2$, where $\int_{-T}^{T} \ell_b^2$ is bounded for $b_0/2 \le b \le 2b_0$. Finally, $\int w_p \Delta \to \int W(\ell_c - \ell_{b_0})$. To see this, write $\Delta_0 = \ell_c - \ell_{b_0}$. Since $|w_p - W| \le 1$, we have $\int |w_p - W||\Delta_0| \le \eta \|w_p - W\|_{L^1} + \int_{|\Delta_0|>\eta} |\Delta_0|$, which tends to 0 as $k \to \infty$ and then $\eta \to \infty$. Moreover, $\int w_p |\ell_b - \ell_{b_0}| \to 0$, since $\ell_b - \ell_{b_0}$ tends to 0 in $L^1_{\mathrm{loc}}(\mathbb{R})$ and uniformly for large $|\tau|$, and J_p is bounded. Letting $k \to \infty$ in (11), we obtain $\mathcal{I}(c) - \mathcal{I}(b_0) = \int W \Delta_0 \ge 0$. □

The last lemma collects two properties of $\mathcal{I}$. The first excludes minimizers near the origin, and the second will rule out that $\mathcal{I}$ has infinitely many minimizers.

Lemma 11. (a) *$\mathcal{I}(c) \to +\infty$ as $c \to 0^+$.*

(b) *Suppose that $0 < \beta < \infty$ and that $W = G$ almost everywhere on $(-\beta, \beta)$, where G is holomorphic in the disc $|z| < \beta$. Then $\mathcal{I}$ is real analytic on $(0, \beta)$.*

Proof. (a) Let $0 < c < 1/2$. Since $\int W \cdot (-2\log c) = -2J_0 \log c$, it suffices to bound $\int W(\log|c^2 - \tau^2| - \log|1 - \tau^2|)$ from below, uniformly in c. On $|\tau| \le 1$, the first logarithm is nonpositive, the second is nonpositive, and $W \le 1$, so this part is at least $\int_{-1}^{1} \log|c^2 - \tau^2| \,\mathrm{d}\tau \ge 2\int_{-1}^{1} \log|u| \,\mathrm{d}u = -4$. On $|\tau| \ge 2$ the difference of the logarithms is nonnegative. On $1 \le |\tau| \le 2$ it is at least $\log(3/4) - \log|1 - \tau^2|$, which is integrable.

(b) Let $0 < \beta' < \beta$. Since W is even, so is G. For $0 < c < \beta'$ we have

$$\mathcal{I}(c) = -2J_0 \log c + 2\int_{-\beta'}^{\beta'} G(\tau) \log|\tau - c| \,\mathrm{d}\tau + \int_{|\tau|\ge\beta'} W(\tau) \log\Big(1 - \frac{c^2}{\tau^2}\Big) \,\mathrm{d}\tau + \text{const.}$$

Here we used $\log|c^2 - \tau^2| = \log\tau^2 + \log(1 - c^2/\tau^2)$ for $|\tau| \ge \beta'$, and the identity $\int_{-\beta'}^{\beta'} G(\tau)\log|\tau + c| \,\mathrm{d}\tau = \int_{-\beta'}^{\beta'} G(\tau)\log|\tau - c| \,\mathrm{d}\tau$. The third term is holomorphic in $|c| < \beta'$, since W is integrable and the logarithm is holomorphic and locally bounded there, uniformly in $|\tau| \ge \beta'$. For the second term, let Log be the branch of the logarithm on $\mathbb{C} \setminus \{-iy : \ y \ge 0\}$ which is real on $(0, \infty)$, and let Γ be the arc $z = \beta' e^{i\theta}$, with θ from π to 0. For real c with $|c| < \beta'$, Cauchy's theorem in the upper half-disc gives $\int_{-\beta'}^{\beta'} G(\tau)\operatorname{Log}(\tau - c) \,\mathrm{d}\tau = H(c)$, where $H(c) = \int_\Gamma G(z)\operatorname{Log}(z - c) \,\mathrm{d}z$; the logarithmic singularity at $\tau = c$ is integrable. Since G is real on $\mathbb{R}$, the real part of the left-hand side is $\int_{-\beta'}^{\beta'} G(\tau)\log|\tau - c| \,\mathrm{d}\tau$. The function H is

holomorphic in a complex neighbourhood of $(-\beta', \beta')$, because $z - c$ stays off the branch cut for z on Γ. Hence $\operatorname{Re} H$ is real analytic on $(-\beta', \beta')$. Since $\beta' < \beta$ was arbitrary, this proves (b). □

We are now ready to prove Theorem 5.

Proof of Theorem 5. By Lemmas 6 and 8 and Corollary 1, there are $p_2 > 0$ and $0 < \kappa \le \kappa' < \infty$ such that b is continuous on $(0, p_2)$ with values in $[\kappa, \kappa']$. The set of limit points of $b(p)$ as $p \to 0^+$ is

$$\mathcal{L} = \bigcap_{0<\delta<p_2} \overline{b((0,\delta))},$$

a decreasing intersection of compact connected sets. Hence $\mathcal{L} = [\alpha, \beta]$ with $\kappa \le \alpha \le \beta \le \kappa'$, and the limit exists if and only if $\alpha = \beta$.

Suppose that $\alpha < \beta$. Choose $p_k \to 0^+$ with $b(p_k) \to \beta$, and pass to a subsequence along which $\mu_{p_k} \to \mu$ vaguely. For $0 < c' < \beta$ we have $b(p_k) > c'$ for large k, so $\mu_{p_k}((0, c')) = 0$. Since $(0, c')$ is open, $\mu((0, c')) = 0$, and therefore $\mu((0, \beta)) = 0$. Hence

$$F(z) = \int_{[\beta,\infty)} \operatorname{Log}\left(1 - \frac{z^2}{a^2}\right) \mathrm{d}\mu(a)$$

is holomorphic in $|z| < \beta$, since the integral converges locally uniformly there by (9), and $\operatorname{Re} F = U_\mu$ on $(-\beta, \beta)$. By Lemma 9, $W = e^F$ almost everywhere on $(-\beta, \beta)$, so $\mathcal{I}$ is real analytic on $(0, \beta)$ by Lemma 11 (b). By Lemma 10, every point of $[\alpha, \beta]$ is a minimum point of $\mathcal{I}$. Hence $\mathcal{I}$ is constant on $[\alpha, \beta]$, and therefore on $(0, \beta)$. This contradicts Lemma 11 (a). Hence $\alpha = \beta$, and $\lim_{p\to 0^+} p\, t_1(p) = \alpha$ lies in $[\kappa, \kappa']$. □

7. Concluding remarks

The uniqueness assumption in Theorem 5. The assumption is used twice: for the evenness of φ_p, which gives the product formula and the bound $t_n \ge n/e$, and in Lemma 6. Lemma 7 holds for any choice of solutions of (1), so the limit profile W does not depend on the choice. Continuity of b is used only to know that the set of limit points of $p\, t_1(p)$ is an interval.

The value of the limit. The proof shows that the limit $b_* = \lim_{p\to 0^+} p\, t_1(p)$ minimizes $\mathcal{I}$ and that every vague limit of μ_p vanishes on $(0, b_*)$. It does not show that b_* is the infimum of the support of such a limit, and it gives no formula for b_*. The limit profile W is tied to the constant c_0 of (B6) through $\int_{\mathbb{R}} W = 2/c_0$, and c_0 is not known in closed form [3, Theorem 1.3].

Problem 1. *Determine* $\lim_{p\to 0^+} p\, t_1(p)$. *Does the limit exist without the assumption that the solution of* (1) *is unique for small* p*?*

The separation constant. Theorem 1 gives no information on how $\sigma(\mathscr{Z}(\varphi))$ depends on p, since the proof is by contradiction. For this reason the exponent $1 + 4/p$ in Lemma 4, which comes from the bound $|I_n|^{2p-1} \leq \sigma^{2p-1}$, carries a constant which depends on the solution φ. A quantitative form of Lemma 3 and Theorem 1 would make this constant explicit.

Acknowledgements. AI assistance was used in developing and checking the arguments and in preparing the exposition; the author has verified the proofs.

References

[1] Andriy Bondarenko, Joaquim Ortega-Cerdà, Danylo Radchenko, and Kristian Seip, *The Hörmander–Bernhardsson extremal function: a preliminary study*, Recent advances in approximation and potential theory—in honor of Edward Saff, 2026, pp. 77–88, DOI 10.1007/978-3-031-95551-8_4. MR5028029 ↑3

[2] ———, *The Hörmander–Bernhardsson extremal function*, Acta Math., to appear, available at `arXiv:2504.05205`. ↑3

[3] Ole Fredrik Brevig, Andrés Chirre, Joaquim Ortega-Cerdà, and Kristian Seip, *Point evaluation in Paley–Wiener spaces*, J. Anal. Math. **153** (2024), no. 2, 595–670, DOI 10.1007/s11854-024-0338-z. Numbering of results as in arXiv:2210.13922v4. MR4802989 ↑1, 2, 3, 4, 5, 10, 11, 12, 13, 14, 15, 19

[4] B. Ya. Levin, *Lectures on entire functions*, Translations of Mathematical Monographs, vol. 150, American Mathematical Society, Providence, RI, 1996. MR1400006 ↑4

[5] Michel Plancherel and George Pólya, *Fonctions entières et intégrales de Fourier multiples*, Comment. Math. Helv. **10** (1937), 110–163, DOI 10.1007/BF01214286. MR1509570 ↑4

[6] Øistein Søvik, *Remarks on three conjectures on point evaluation in Paley–Wiener spaces* (2026), preprint. Available at `https://github.com/Oisov/math-preprints`. ↑3, 13

University of Oslo, Oslo, Norway
Email address: `oistein.sovik@usit.uio.no`